PhD IO
Problem Set 1
Due November 16

Form a team to do this homework (teams can have 2 or 3 members – if you want to work on this alone, you may) – or you can do it by yourself (although I recommend doing this with a group). On the Canvas site there is a Monte Carlo generated Excel data set containing information on the outcomes in 60 health insurance markets. The data contain firm/market level information on health plan characteristics, premiums (measured in $1,000) and shares (which are calculated accounting for the outside good share). The number of insurers participating in these markets varies but insurers only offer one product in each market although they participate in multiple markets. There are three plan characteristics: Health Maintenance Organization (HMO) $\in \{0, 1\}$ which many consumer dislike, the ratio of providers that are in-network (takes the values between 0 and 1), and the mean customer satisfaction score (takes the values between 0 and 1). Preferences are stable across the markets. The outside good is no health insurance – i.e. uninsurance. Please submit your answers on Canvas.

With these data, answer the following questions. Your answers should include a 4-to-5 page write-up discussing the results. The use of outside programs (e.g. PyBLP) and AI agents is allowed although you will learn more by writing your own programs. Your answers should include a description of which programs and AI agents you used and a description of how you used them in your analysis. You do not need to submit the actual programs.

1. What are the utility parameters? Estimate the parameters of the utility function using a 1) Logit, 2) a Nested Logit, 3) and using the BLP specifications. That is, for the logit model estimate the parameters of the following utility specification: $u_{ijm} = \xi_{jm} + x_j\beta - \alpha p_{jm} + e_{ijm}$ where x is the characteristics vector, p is premium and e_{ijm} is the logit error. The unobservable quality of the product is given by $\xi_{jm} = \bar{\xi}_j + v_{jm}$ where v_{jm} is an iid error. The nested logit specification has the analogous structure as in Berry (1994). For the Nested Logit specification use the HMO variable to group products. The BLP specification is given by $u_{ijm} = \xi_{jm} + x_j\beta_i - \alpha p_{jm} + e_{ijm}$ where parameters for the product characteristics are assumed to be distributed $N(\bar{\beta}, \sigma^2)$. The price parameter, α, does not vary across consumers. You may want to experiment with different combinations of instruments (e.g. Hausman, BLP, Gandhi/Houde). If you have problems getting BLP to work, reach out to me.

2. What is the mean own price and mean mark-up implied by the different models? How do the mark-ups vary with market structure?

3. What is the relationship between MC and the plan characteristics implied by the BLP model? Note that you could estimate the BLP model using supply side moments which will give you the answer directly but you do not need to use that approach.

4. Suppose the government gives the plans a $250 subsidy for each consumer that signs up

with them. Using your BLP estimates, recompute and report the Nash-Bertrand equilibrium premiums. To do this you will need to solve for the post-subsidy Bertrand-Nash equilibrium using a nonlinear equation solver.

- How much will premiums change? How much will the uninsurance (the outside good) rate change?
- How much will profits per enrollee for the merging and non-merging plans increase?
- How much will mean consumer surplus change? Does it vary by market structure?

5. Suppose Plans 2 and 4 merge and set premiums to maximize joint profits. Using the BLP estimates answer the following:

- How much will premiums change? How much will the uninsurance (the outside good) rate change?
- How much will profits per enrollee increase?
- How much will mean consumer surplus change? Does it vary by market structure?

6. Suppose the merger enables efficiencies of a 5% reduction in each plan's marginal cost. How much will premiums change? Should the antitrust authorities let the merger proceed? Why or why not?